

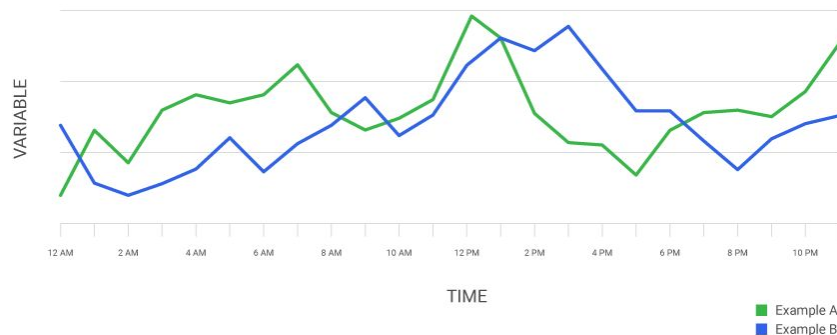
Kaggle - Time Series #09



Basics and Intro

What is time series:

- A set of observations recorded over time (i.e daily/monthly..)



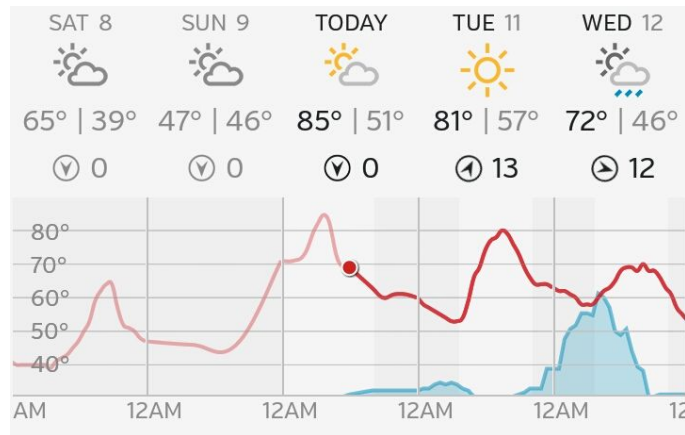
	date	value
1	2023-01-01	145
2	2023-02-01	212
3	2023-04-01	265
4	2023-05-01	299
5	2023-06-01	345
6	2023-09-01	278
7	2023-10-01	256
8	2023-11-01	202
9	2023-12-01	176

Forecasting : predicting a value in the future

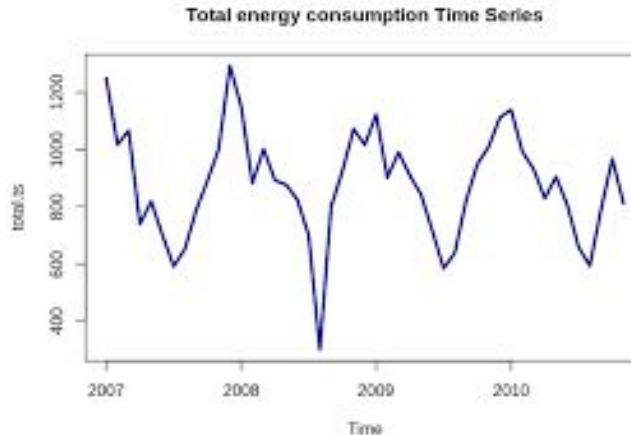
Uses

When would Time series forecasting be useful? Can you think of some examples?

Weather Forecasting



Energy consumption



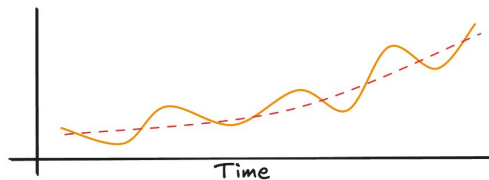
Other
examples:

Sales/
Demand
forecasting

Patterns over time

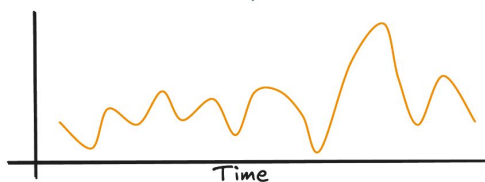
TREND

It is the long-term tendency and shows if data increases or decreases over a long period of time.



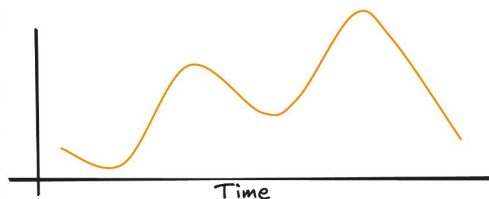
IRREGULAR

Fluctuations that are not regular variations and are purely random or irregular. They are: uncontrollable, unpredictable, and erratic.



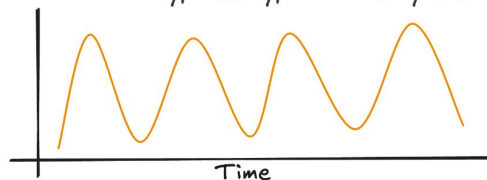
CYCLICAL

Repeating movements or oscillations over more than a year.



SEASONAL

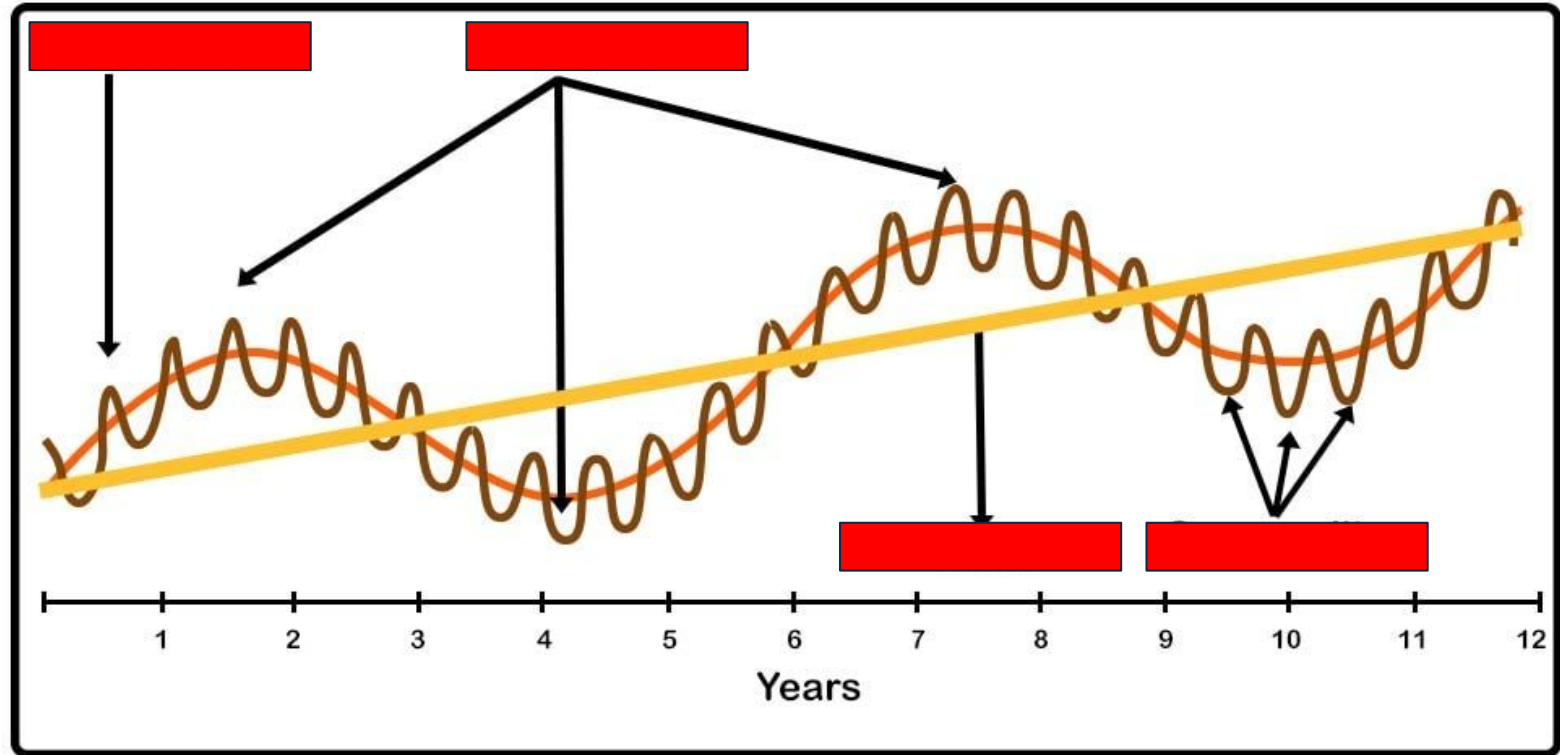
They show consistent behavior over time. These are patterns that repeat regularly within a year such as: daily, weekly, or monthly trends



Periodic Fluctuations:

There are some components in a time series which tend to repeat themselves over a certain period of time.

Patterns over time



Lag Features

What are Lag features:

Features (columns) that represent values of a time series at **previous** time steps , allowing the model to learn patterns over time

- Lag1 = t-1
- Lag2 = t-2 ..

```
df['Lag_1'] = df['NumVehicles'].shift(1)  
df.head()
```

	NumVehicles	Time	Lag_1
Day			
2003-11-01	103536	0	NaN
2003-11-02	92051	1	103536.0
2003-11-03	100795	2	92051.0
2003-11-04	102352	3	100795.0
2003-11-05	106569	4	102352.0

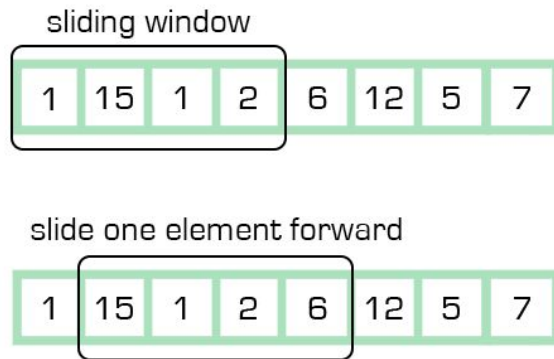
Window Features

What are window features?

They are new variables created by calculating statistics like mean , max or standard deviation over a “sliding” window

Examples:

- Average sales of the previous 3 days
- Max value of the previous 3 months



Extracting Data and Time features

- This is when you break down timestamps into meaningful numerical components (year, month, day hour etc.) to reveal patterns , trends, and seasonality
- For example you may want to capture the relationship of energy usage vs hour of the day. Less energy usage at night , more in the evening / day time

	date	year	month	day	hour	minutes	day of year	day_of_year	week_of_year
0	2019-12-01 01:17:00	2019	12	1	1	17	335	335	48
1	2019-12-02 08:32:00	2019	12	2	8	32	336	336	49
2	2019-12-03 15:47:00	2019	12	3	15	47	337	337	49
3	2019-12-04 23:02:00	2019	12	4	23	2	338	338	49
4	2019-12-06 06:17:00	2019	12	6	6	17	340	340	49

Data preprocessing tips

Convert dates to “datetime” object: this is a specific data type in python as sometimes in a dataset the date can be a string or number etc.

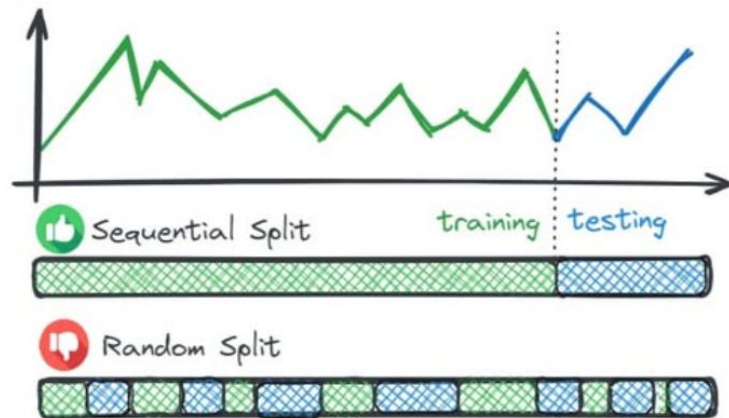
Time series must have consistent regular time intervals (e.g hourly, daily). Missing data should be filled

Time Series splitting

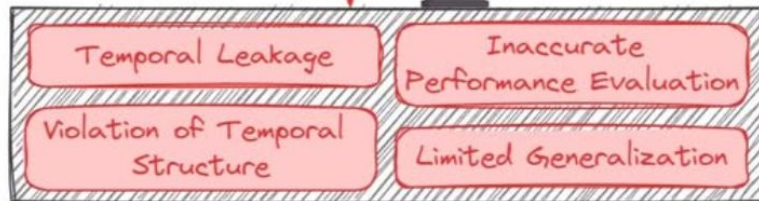
Never use random cross validation : as this messes up the chronological order of time series model / enables **data leakage** (when data can look into the future)

Time series splitting should be in order

Sequential Split in Time Series



leads to



Evaluating the model

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

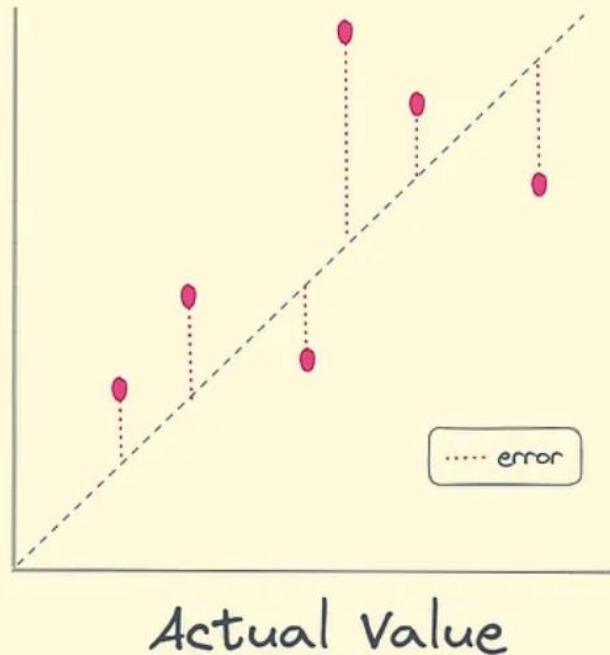
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

~~$$R^2 = 1 - \frac{SSR}{SST} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$~~

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

Predicted Value



Summary

- Seasonality , trends , cyclicity and noise
- Lag features
- Window features
- Extracting date and time features
- Data preprocessing tips - regular intervals
- Data Leakage

Feedback Form

